

Solution Requirements Specification

Smart-Lender Project Documentation Suite

Project Name	Smart Lender - AI-Powered Loan Approval Prediction System
Document Type	Solution Requirements Specification
Project Phase	Phase 2: Requirement Analysis
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Organization / College	Vishnu Institute of Technology
Status / Version	Approved / v1.0 (Exact Replica Format)

1. Functional Requirements Table

ID	Category	Requirement Specifications	Status
FR-1	Authentication	Optional user credential verification portal	Pending
FR-2	Inputs Form	Accept applicant features (Income, Dependents, Credit Score, Property Area)	Completed
FR-3	Data Pipeline	Missing values mapping, scaling parameters normalization via joblib matrices	Completed
FR-4	ML Engine	Assess risk criteria using balanced XGBoost classifier	Completed
FR-5	Reporting UI	Render results pages showing Approved/Rejected with custom layouts	Completed

2. Non-Functional Requirements Table

Parameter	Target Standard Requirement	Validation Result Status
Performance	Return classification output in the user browser in < 2 seconds	Passed (<0.5s)
Scalability	Handle multiple requests without memory leaks	Passed
Reliability	Consistent calculations for duplicate candidates	Passed (Deterministic PKL files)